Welcome to MCS 472

1. About the Course
   - content
   - expectations
2. Julia and the Jupyter notebook
   - installing and running Julia
   - the Jupyter notebook and JupyterLab
3. Statistical Reasoning
   - random variables
   - four problems, four distributions

MCS 472 Lecture 1
Industrial Math & Computation
Jan Verschelde, 12 January 2026

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Catalog Description

MCS 472. Introduction to Industrial Math and Computation.

Technical writing and oral presentations in preparation for industrial projects. Topics include quality control, operations research, cost-benefit analysis, differential equations, using scientific software.

Course Information: Extensive computer use required.
Prerequisite(s):
Grade of C or better in MCS 471 or consent of the instructor.

Recommended background: *Designed for students with a desire to explore mathematics via practical field work.*

MCS 472 is on the computational track in the MCS major, next to

- MCS 320, Introduction to Symbolic Computation; and
- MCS 471, Numerical Analysis.

Content and Organization of the Course

The course is based on the book by Charles R. MacCluer: *Industrial Mathematics. Modeling in Industry, Science, and Government,* Prentice Hall, 2000.

The textbook is optional, not required.

There are three parts:

1. Statistical Reasoning, Data Acquisition and Processing
2. Linear Programming, Regression, Cost-Benefit Analysis, Machine Learning
3. Ordinary Differential Equations, Feedback Control, Modeling with Partial Differential Equations

Each part ends with one week of project presentations.

The course is driven by problems and projects.

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Purpose of the Course

The main goal is to prepare for the solving of industrial problems.

Three aspects:

1. Translate the real-world problems into a mathematical model.
2. Compute a solution using scientific software.
3. Write a technical report and make an in-class oral presentation.

Free and Open Source Software environments:

- GNU Octave
- SageMath and/or numpy, sympy, scipy
- Julia and the Jupyter notebook

`CoCalc`: Collaborative Calculation and Data Science (cocalc.com)

This is a *computational* course, not a programming class.

Evaluation

Your score depends for

- 20% on homework,
- 60% on the projects, and
- 20% on the exams.

By default, unless explicitly stated that collaborations are allowed,

all submitted solutions must be your own work.

Course URL:

`https://homepages.math.uic.edu/~jan/mcs472`

Alternative URL: `https://janverschelde.github.io/mcs472`

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installing and running Julia

The current stable version of Julia is 1.12.4 (January 8, 2025).

Julia is free software. To install, do the following:

1. Visit `https://julialang.org`.
2. Install the `Juliaup` installation manager.
3. Adjust the `path` environment variable if needed.

Two ways to run Julia:

1. in an interactive Terminal or PowerShell window; or
2. at the command prompt, type

   ```
   julia program.jl
   ```

   where `program.jl` is a Julia program.

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the Jupyter notebook and JupyterLab

The Jupyter notebook is a web based interface for computations.

- Cells can contain code or text.
- A well structured notebook can be printed as a technical report.

JupyterLab is the latest notebook environment.

To install the notebook with Julia, do the following:

1. In a Julia session, type `using IJulia`.
2. Answer `y` to download.
3. Type `using Pkg; Pkg.build("IJulia")`.
4. Type `using IJulia`.
5. Type `notebook()`.

The first time, the system will prompt you to install Jupyter.

Exercise 0: Install Julia and the Jupyter notebook on your computer.

This is the most important exercise of this week!

installing packages

It is not good practice to install packages in a Jupyter notebook.

Instead, open a terminal lauching the Julia prompt and type

```
using Plots
```

to install the `Plots` package.

In this lecture, we will also use the package `QuadGK`.

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random variables

A random variable X models an experiment.

- $P(X = x)$ is the probability that X takes the value x.
- $F(X \leq x)$ is the probability that X takes a value less than x.

F is the cumulative probability distribution function:

$$P(a \leq X \leq b) = F(b) - F(a) = \int_a^b dF.$$

If F is differentiable, then $f = F'$ is the density function.

expected value and variance

A random variable X with cumulative probability density function F has *expected value*

$$\mu = E[X] = \int x dF$$

and *variance*

$$\sigma^2 = E[(X - \mu)^2].$$

The σ is *the standard deviation* of X.

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the Rolling Dice Problem

the discrete uniform distribution

We roll a fair 6-sided die ten times.

- What is the expected value of the sum of the outcomes?
- What is the variation in the expected value?

the discrete uniform distribution

In rolling a fair 6-sided die,
every side has equal probability of $\frac{1}{6}$.

- The mean value of one throw is 7/2, computed as

$$\mu = 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 5\left(\frac{1}{6}\right) + 6\left(\frac{1}{6}\right) = 21/6.$$

  This implies that after 10 rolls, we may expect the sum to be 35.

- The variance is $35/12 \approx 2.9166$, computed as

$$\sigma^2 = \frac{1}{6}\sum_{i=1}^{6}\left(i - \frac{7}{2}\right)^2.$$

  For one roll, we thus have 3.5 ± 2.9.
  For ten rolls, we expect 35 ± 29.

rolling a pair of dice

Exercise 1:

In Julia, `rand((1,2,3,4,5,6))` simulates the roll of a 6-sided die.

Run 1000 simulations of rolling a pair of 6-sided dice.

1. How many times did $2, 3, 4, \ldots, 12$ occur?
   Make a histogram of the sum of the outcomes.
2. Graph the cumulative distribution function $F(x)$.

The Life Span Problem

the normal distribution

Suppose a light bulb has an average life span of 1000 hours,

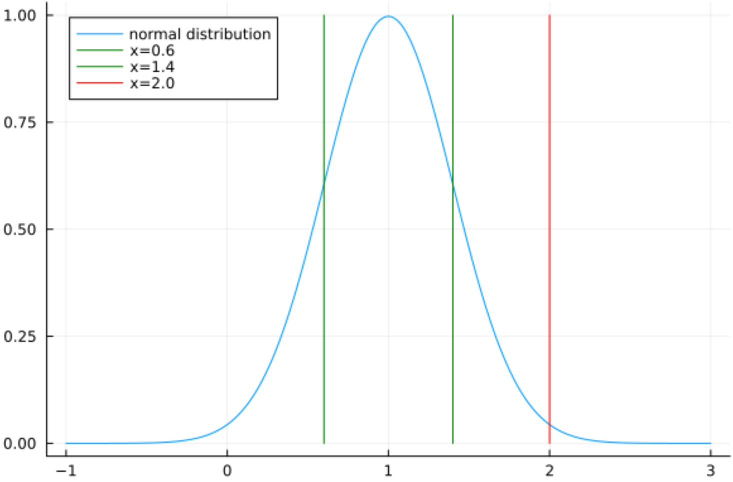
with a standard deviation of 400 hours.

Assuming a normal distribution,

what is the probability that a bulb lasts 2000 hours or longer?

applying the normal distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}, \text{ for } \mu = 1, \sigma = 0.4$$



using `QuadGK`

$$P(X > 2) = \int_2^\infty \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2} \quad \text{for } \mu = 1, \sigma = 0.4$$

With `QuadGK` in a Julia session:

```
julia> using QuadGK

julia> f(x) = (1/(0.4*sqrt(2*pi)))*exp(-(x-1)^2/(2*0.4^2))
f (generic function with 1 method)

julia> val, err = quadgk(f, 0.6, 1.4, rtol=1.0e-8)
(0.682689492137086, 9.098977127308672e-11)

julia> val, err = quadgk(f, 2.0, 100.0, rtol=1.0e-8)
(0.006209665325776137, 2.981577782577757e-12)
```

The probability that a light bulb last 2000 hours or longer is 0.0062.

on planning a trip ...

Exercise 2:
Your friend asks your advice on a planned motor bike trip of 1,000 miles through the desert.

Because of the rough terrain, tires are expected to last 1,500 miles with a standard deviation of 300 miles.

As soon as one tire expires, the trip is over.

What is the probability your friend gets stranded in the desert?

The Staffing Problem

the binomial distribution

On average, for every three staffers who are emailed for a shift, only one shows up.

How many staffers must be emailed to guarantee at least 20 show up, with a confidence of 90%?

applying the binomial distribution

On average, for every three staffers who are emailed for a shift, only one shows up. How many staffers must be emailed to guarantee at least 20 show up, with a confidence of 90%?

Observe: $1/3 = p$ is the probability of a presence. Of n emails sent, the probability that k are present is $p(k) = \left(\begin{array}{c} n \\ k \end{array} \right) p^k (1-p)^k$, where the factor $\left(\begin{array}{c} n \\ k \end{array} \right)$ equals all subsets of size k out of a total of n.

Find n so that the probability that less than 20 are present $< 10\%$.

$$\sum_{k=0}^{19} p(k) = \sum_{k=0}^{19} \left(\begin{array}{c} n \\ k \end{array} \right) \left(\frac{1}{3} \right)^k \left(\frac{2}{3} \right)^{n-k} < 0.1$$

a command line solution

Using Julia as a calculator to evaluate $\displaystyle\sum_{k=0}^{19} p(k) = \sum_{k=0}^{19} \binom{n}{k} \left(\frac{1}{3}\right)^k \left(\frac{2}{3}\right)^{n-k}$

```
julia> c(n,k) = factorial(n)/(factorial(k)*factorial(n-k))
c (generic function with 1 method)

julia> f(n) = sum([c(big(n),big(k))*(1/3)^k*(2/3)^(n-k) for k=0:19]
f (generic function with 1 method)

julia> f(60)
0.45158877578957089188367551591955099133984421265292918618121082716

julia> f(73)
0.11365054893565017115884506290529067137584443424671088652372870738

julia> f(74)
0.09950717529629137197668084287380332801796993918620879724884986272
```

At least 74 emails must be sent to have 20 present with 90% confidence.

an overbooking strategy for an airline

Exercise 3:

Suppose, on average, five percent of travelers do not show up or arrive too late to board the air plane.

A flight is oversold if there are more tickets sold than the number of available seats on the plane.

What is the percentage of the number of available seats that can be oversold

and still give every traveler who shows up on time a seat?

The Arrival Problem

the Poisson distribution

A market has 60 customers per hour on average.

What is the probability that more than 20 customers check out in one quarter hour?

applying the Poisson distribution

A market has 60 customers per hour on average.
What is the probability that more than 20 customers check out in one quarter hour?

Observe: time is continuous. Time unit = one quarter of an hour.
The random variable X is the number of arrivals per quarter hour.
On average: 15 customers per quarter hour.

The Poisson distribution has parameter λ and average $\mu = \lambda T$, where T is the time unit.

$$p(k) = \frac{(\lambda T)^k}{k!} e^{-\lambda T} \quad F(x) = e^{-\lambda T} \sum_{k=0}^{x} \frac{(\lambda T)^k}{k!}$$

$F(x) = P(X \leq x) = 1 - P(X > x)$, so compute $e^{-15} \displaystyle\sum_{k=0}^{20} \frac{15^k}{k!}$.

a Julia function

```
"""
    A market has 60 customers per hour on average.
What is the probability that more than 20 customers
check out in one quarter hour?
"""
function arrival()
    q = 0           # sum of probabilities
    m = 20          # number of arrivals
    lT = 15         # arrivals per quarter hour is lambda*T
    p = 1           # accumulates the product
    for k=1:m
        p = p*lT/k  # update the product
        q = q + p   # update the sum
    end
    q = q*exp(-lT)  # normalization
    p = 1 - q
    println("probability : ", p)
end
```

Running `arrival()` prints `probability :  0.08297121593378054`.

on the numerical evaluation of a formula

Exercise 4:

Consider evaluating $e^{-15} \sum_{k=0}^{20} \frac{15^k}{k!}$ with an array comprehension.

1. Give the code for the array comprehension in Julia, with the default 64-bit floating-point arithmetic.
2. Explain why the numerical value of the comprehension differs from the output of the function `arrival`.
3. Use computer algebra to compute the exact value.

available through our UIC library

For more on distributions, consider: *Statistical Distributions*

by Catherine Forber, Merran Evans, Nicholas Hastings,

and Brian Peacock, Fourth edition, John Wiley & Sons, 2011.

This handbook provides a concise summary of 40 major distributions.

Examples are mentioned for each distribution.